

Environmental Adaptation Modeling: Predicting Alien DNA and Protein Sequences Through Chemical Environment Simulation

1 Background/Motivation

Life on Earth evolved under specific environmental constraints (atmospheric composition, mineral availability, chemical gradients) that shaped genetic codes and protein structures. Exoplanets present vastly different chemical environments with varying trace metal concentrations and atmospheric compositions. Current genomic AI models, trained exclusively on Earth organisms, cannot predict how life might adapt to alien environments, limiting our understanding of astrobiology’s fundamental principles.

2 Problem Formulation

How can we develop an AI model that generates plausible DNA and protein sequences for organisms adapted to specific alien environmental conditions, and predict their resulting biological capabilities?

3 Data & Methods

Data Sources: [UniProt](#) protein database (>200M sequences), [NCBI GenBank](#), extremophile environmental metadata from [GOLD](#) (Genomes OnLine Database), [NASA Exoplanet Archive](#) data (~5,000 exoplanets), and meteorite geochemical data from the [Meteoritical Bulletin Database](#) for realistic alien environmental parameters.

Model Architecture: Two-stage conditional generative model: (1) Environment-to-sequence transformer taking chemical concentration vectors (C, N, O, S, P, Si, Fe, etc.) as input to generate DNA/protein sequences, (2) Sequence-to-function predictor using protein language models ([ESM-2](#)) to predict structural and functional properties. Input: 10–20 dimensional environmental vectors. Output: nucleotide/amino acid sequences plus predicted capabilities.

Training Strategy: Establish correlations between Earth’s environmental gradients and extremophile adaptations (hydrothermal vents, alkaline lakes, etc.) to learn environment-genome relationships, then extrapolate to alien conditions beyond Earth’s parameter ranges.

Validation: Evaluate by predicting known extremophile adaptations, measuring sequence diversity/biological plausibility, and structural analysis using [AlphaFold](#)-style tools. Implementation via [PyTorch](#) with [Hugging Face transformers](#) on university computing clusters.